

Power Series Solution: Bessel's Equation

The differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0 \rightarrow (1)$$

is called Bessel's equation of order 'n' and its particular solutions are called Bessel functions of order n.

Solution Here $x=0$ is a regular singularity

so we can take the solution as

$$y = x^m (a_0 + a_1 x + a_2 x^2 + \dots) = \sum_{k=0}^{\infty} a_k x^{m+k}$$

The complete solution of (1) is

$$\boxed{y = C_1 J_n(x) + C_2 J_{-n}(x)} \rightarrow (*)$$

where

$$J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4 (2n+2)(2n+4)} - \dots \right]$$

$$\sim J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \sum_{k=0}^{\infty} \frac{(-1)^k \Gamma(n+1)}{2^{2k} \cdot k! \Gamma(n+k+1)} x^{2k}$$

$$\sim \boxed{J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{n+2k}} \rightarrow \textcircled{A}$$

which is called Bessel function of the first kind of order n .

$$\text{and } \boxed{J_{-n}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(-n+k+1)} \left(\frac{x}{2}\right)^{-n+2k}} \rightarrow \textcircled{B}$$

which is called Bessel function of the first kind of order $-n$

(P.S. The proof is not required in exam. But you go through the solution. Recall the expression $J_n(x)$ & $J_{-n}(x)$)